

Windows Server 2025: the beginners home lab guide



Index

Page 2: Minimum requirements and server setup wizard

Page 3: User Management and client configuration

Page 4: Server snapshots

Page 6: Host snapshots

On July 7th, 2026, I've deployed a Windows Server virtualized lab environment mastering my Active Directory abilities, a fundamental skill required for most IT roles all around the world. In this report, you'll find a complete guide to setting up a working Windows Server home lab at home!

Required software and files: VMware Workstation Pro, Windows 11 and Server 2025 ISO files

Minimum requirements: 16 GB of RAM, otherwise your host will stutter with 2 or more VMs open at the same time.

Procedure for Windows Server

1. Install VMware Workstation Pro, Windows 11 and Server 2025 from their official websites.
2. Complete the **"New Virtual Machine Wizard"** on your virtual machine, choosing the best network adapter setting for you (**NAT: able to connect to the internet, Host Only: private network with no internet access**, better option for isolated environments).
Assign at least 4GB of RAM. If you go lower, the OS is going to be very slow!!!
3. When the Windows setup is done, go to the **Network and Sharing Options** in the control panel and change the network adapter configuration based on the settings mentioned in step 2. In my case (NAT), you need to open **Virtual Network Editor** and go to **VMnet8 → DHCP Settings** and choose one address from the pool in **Starting IP Address**. **This is going to be your static IP address, the one you'll use as a DNS on your client once the configuration is done.** The **Default Gateway** is shown on **"Gateway IP"** by clicking to **NAT Settings** in Virtual Network Editor. For the DNS Settings, write the following address **127.0.0.1**. (**configure these options BEFORE the Active Directory wizard!!!**)
4. Once you have changed your network settings the server **will not be able to access Internet**, don't panic since it's completely normal for **security purposes** (imagine that our virtual environment is the real company's server, if it's exposed to the external network it may encounter to serious threats like ransomware able to compromise every employee's info and work!)
5. Open **Server Manager** and click Next until you are at the **Server Roles**: check the **Active Directory Domain Services** and click Install.
6. Once you installed the ADDS, a yellow flag will appear on the screen, click on it to **Promote this server to a domain controller**.
7. **Add a new forest**: here you are going to create a new domain (**must finish with a .local/.lan extension**) and a DSRM password (**writing it somewhere safe is suggested**). If everything is done well, you won't encounter any problem and be ready to restart.
8. After the restart, log in to the server with the domain name (DOMAIN-NAME\Administrator) and verify that the Active Directory Domain Services are active in Server Manager

Setting up user management and privileges

Users:

1. Go to Server Manager (this program will run automatically at every login) → Tools → Active Directory Users and Computers
2. Right click on the domain name → New → Organizational Unit and rename it if you want.
3. Right click on the Organizational Unit you've just created → New → User

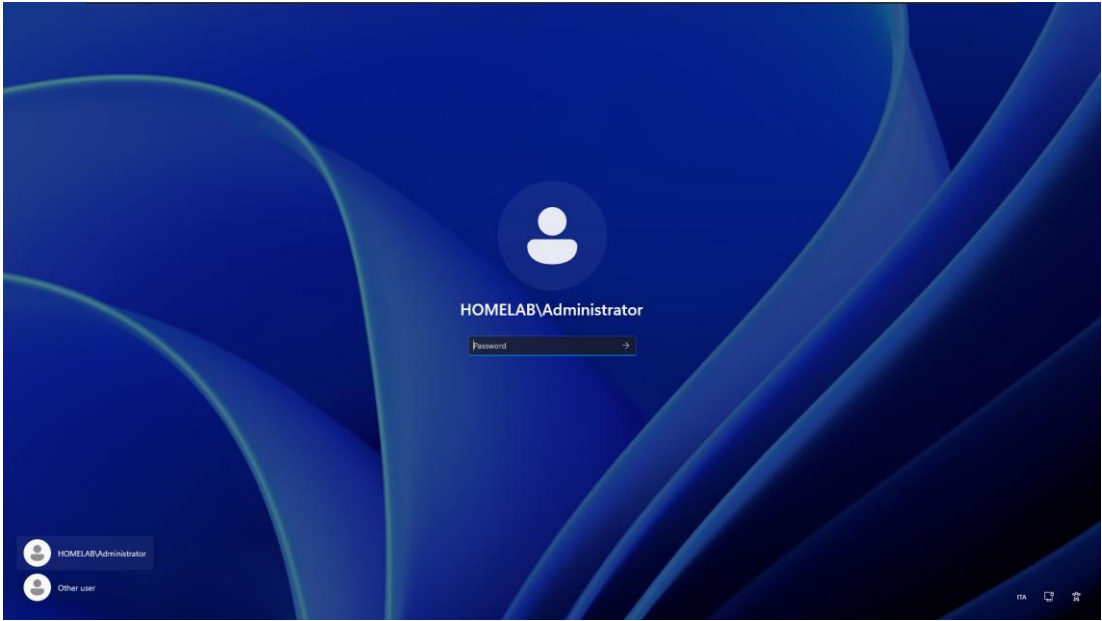
Privileges (such as UAC strength, password's length, default wallpaper, etc.):

1. Go to Server Manager → Tools → Group Policy Management
2. Right click on the organizational unit you've created before → Create a GPO in this domain and link it here... and rename it to not confuse them (UAC for one, Wallpaper for the 2nd, etc...)

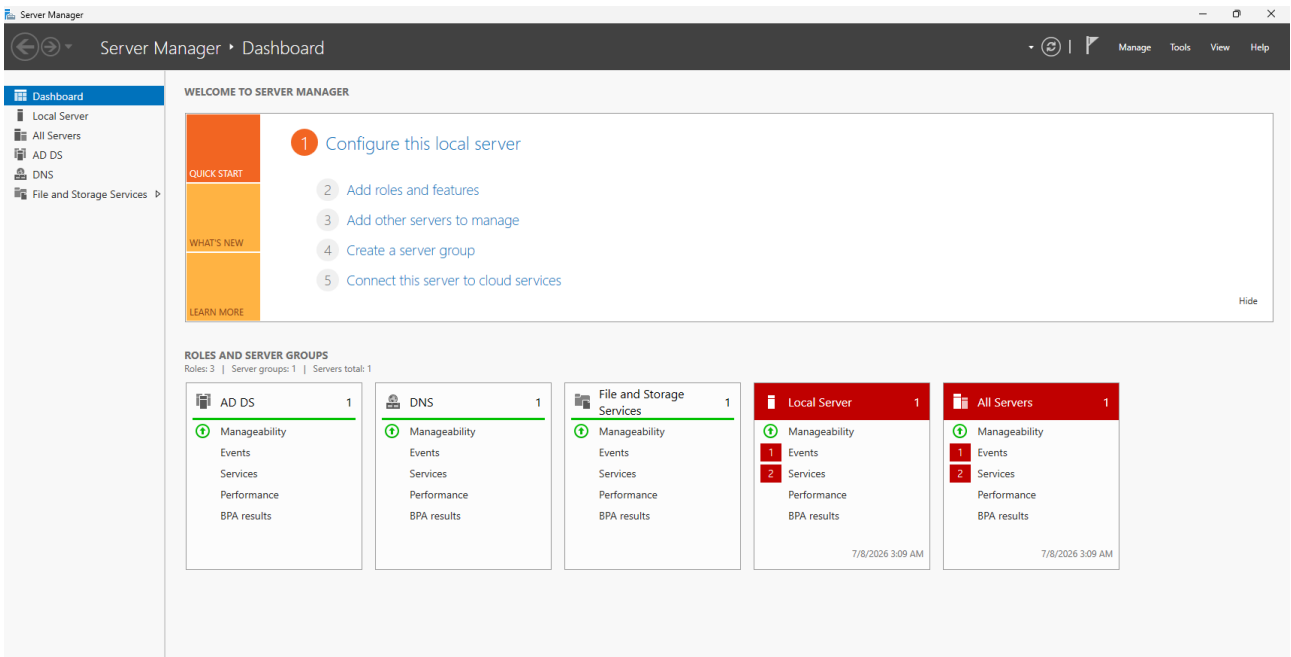
Joining to domain

1. First, launch your Windows Client VM and log in as you would do normally
2. Click on the search/Windows logo and write "Join a domain" and log in with the credentials made in the server VM.
3. If it's all right, a DOMAIN-NAME/Username user will pop up at every logon.

Snapshots (Server)



Windows Server 2025 lock screen after the Active Directory config.



Service Manager

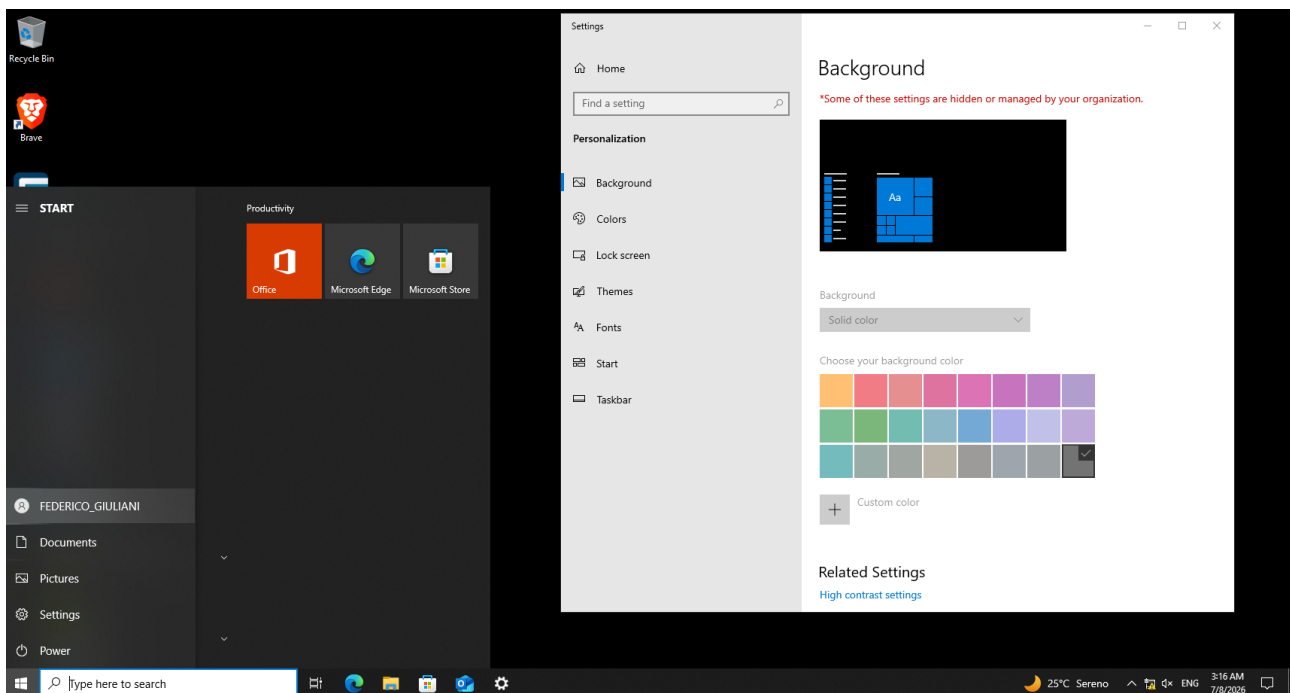
Name	Type	Description
 FEDERICO GIULIANI	User	

Active Directory Users and Computers with one user in the “Utenti Base” OU (Organizational Unit)

Utenti Base							
Linked Group Policy Objects		Group Policy Inheritance		Delegation			
Link Order	GPO	Enforced	Link Enabled	GPO Status	WMI Filter	Modified	Domain
1	 Blocco	No	Yes	Enabled	None	7/8/2026...	homelab...
2	 UAC	No	Yes	Enabled	None	7/8/2026...	homelab...
3	 Sfondo	No	Yes	Enabled	None	7/8/2026...	homelab...

GPM (Group Policy Management) with 3 GPOs

Snapshots (Windows 10 client)



Windows 10 after joining in the domain (not allowed to change the wallpaper, no network access caused by the server shut down)

```
C:\Windows\system32\cmd.exe
C:\Users\FEDERICOGIULIANI>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-I9VCA48
Primary Dns Suffix . . . . . : homelab.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : homelab.local
                                localdomain

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . : localdomain
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-5A-B5-26
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.136.131(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, July 8, 2026 3:15:58 AM
Lease Expires . . . . . : Wednesday, July 8, 2026 3:45:57 AM
Default Gateway . . . . . : 192.168.136.2
DHCP Server . . . . . : 192.168.136.254
DNS Servers . . . . . : 192.168.136.128
Primary WINS Server . . . . . : 192.168.136.2
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\FEDERICOGIULIANI>
```

Host network information in a domain (the DHCP server is the server's IP address, and the DNS is the domain name)